

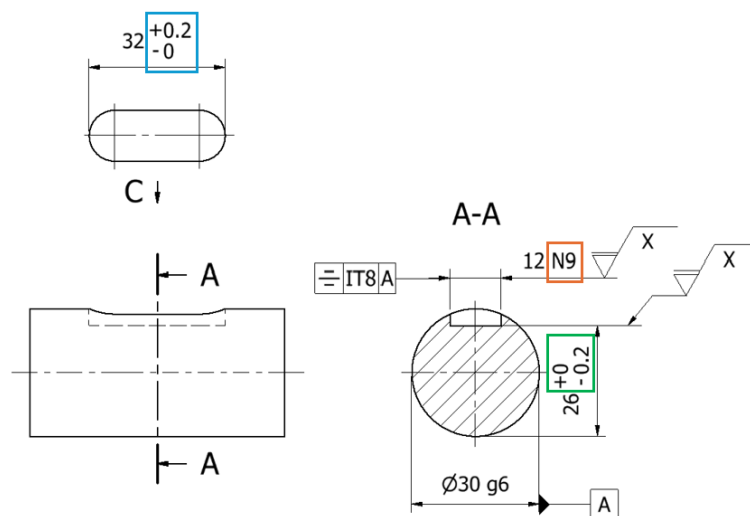
SHAFT DETAILING GUIDE

Views

- ☐ Side Orthographic View of Shaft
- ☐ Sectional View of Keyway
- ☐ Cropped Auxiliary View of keyway
- ☐ Detail Views

Primary Features

Parallel Key Keyways (& General Shaft Fittings (Gears, etc.))



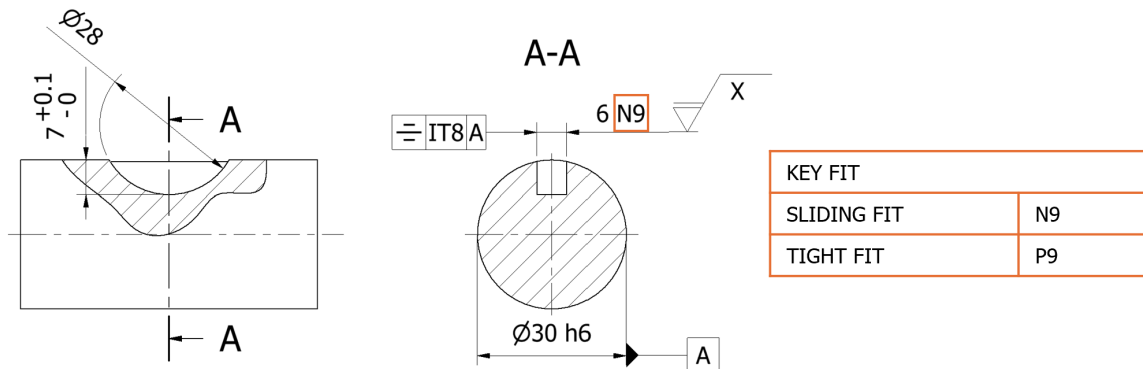
KEY FIT	
SLIDING FIT	N9
TIGHT FIT	P9

KEYWAY DEPTH	
SHAFT DIAMETER	TOL
<= 22	0.1
> 22 (UP TO 150)	0.2

KEYWAY LENGTH	
NOMINAL KEY LENGTH	TOL
< 28	0.2
29 TO 80	0.3
80 TO 250	0.5

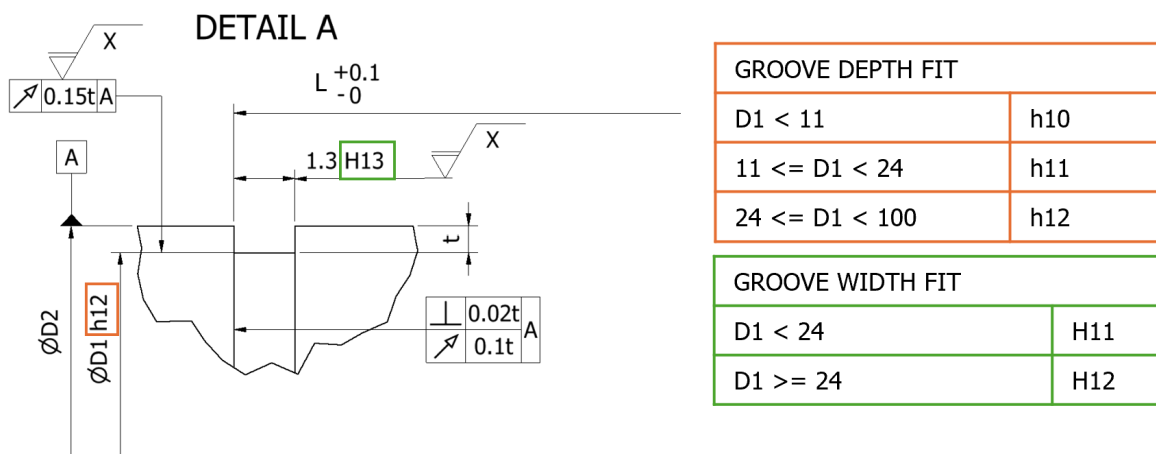
- ☐ Main Nominal Diameter g6 (and Datum)
- ☐ Keyway Depth Tolerance
- ☐ Keyway Fit Tolerance
- ☐ Keyway Length Tolerance
- ☐ Symmetry Geometric Tolerance
- ☐ Surface Finish

Woodruff Key Keyways (& General Shaft Fittings (Gears, etc.))



- ☐ Main Nominal Diameter h6 (and Datum)
- ☐ Nominal Keyway Diameter Dimension
- ☐ Keyway Depth Tolerance
- ☐ Keyway Width Fit
- ☐ Symmetry Geometric Tolerance
- ☐ Surface Finish

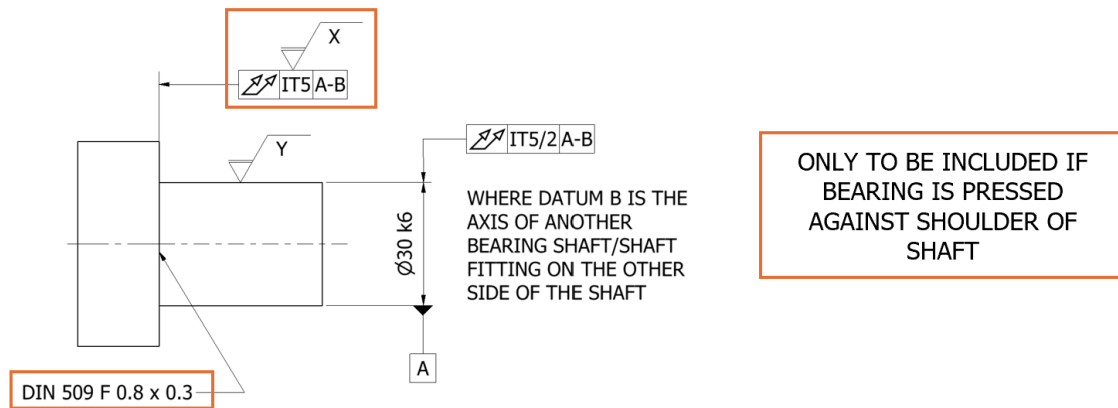
External Circlips



- ☐ Groove Depth Fit (on main orthographic view)
- ☐ Geometric Tolerances (factor * t)

- ☐ Groove Width Fit
- ☐ Datum
- ☐ Surface Finish
- ☐ Position of Circlip Groove (on main orthographic view)

Deep Groove Ball Bearings

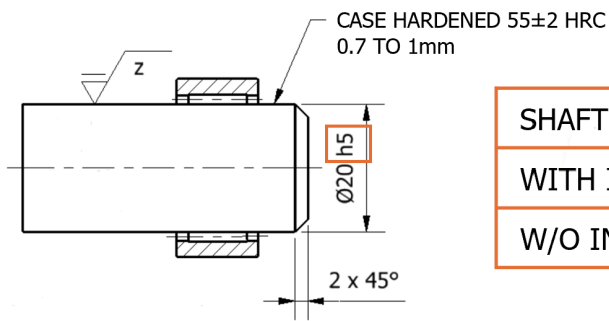


- ☐ Nominal Shaft Diameter k6 fit (with Datum)
- ☐ Total Runout IT5/2 Geometric Tolerance
- ☐ Surface Finish (Y)

IF PRESSED AGAINST SHOULDER,

- ☐ DIN 509 Undercut
- ☐ Total Runout IT5 Geometric Tolerance
- ☐ Surface Finish (X)

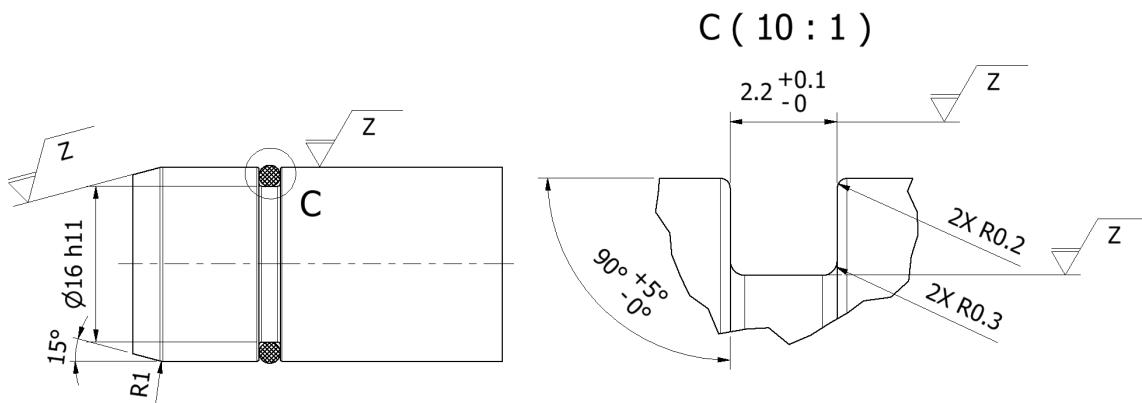
Needle Bearings



SHAFT DIAMETER FIT	
WITH INNER RACE	k5
W/O INNER RACE	h5

- ☐ Shaft Diameter Fit (k5/h5)
- ☐ Chamfer
- ☐ Case Hardening
- ☐ Surface Finish (Z)

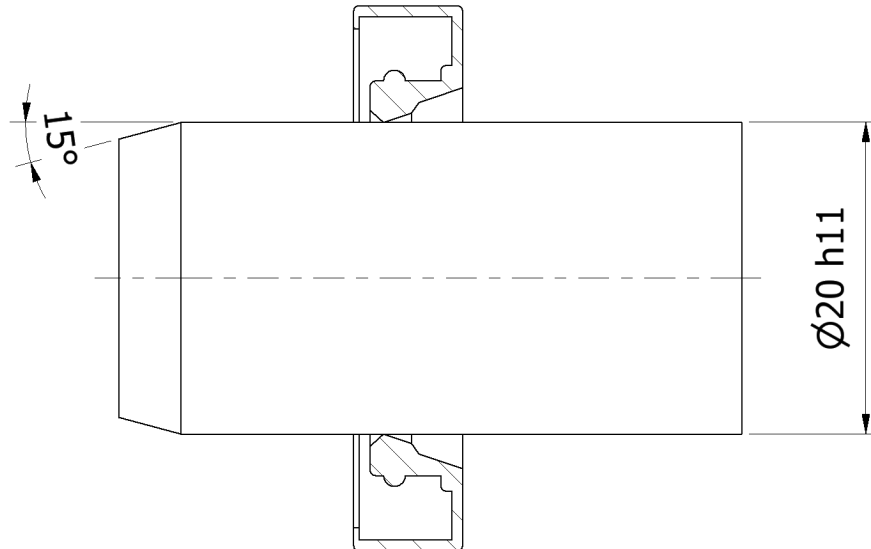
O-Rings



- ☐ Groove Diameter h11
- ☐ 15 deg chamfer with R1 Fillet (chamfer amount is non-critical, pick suitable value)
- ☐ Groove Width Tolerance (+0.1)
- ☐ Angle Tolerance (+5deg)
- ☐ 2 X R0.2 and 2 X R0.3

- ☐ Surface Finish (Z)

Lip Seals

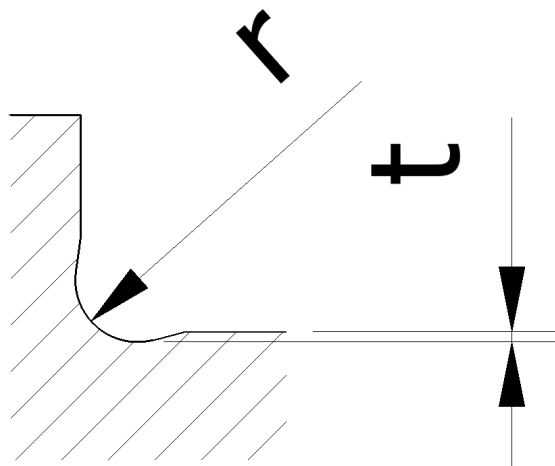


- ☐ Shaft diameter h11
- ☐ 15 deg chamfer (chamfer amount is non-critical, pick suitable value)
- ☐ **SURFACE FINISH Z (NOT IN DWG OOPS)**

Undercuts

All perpendicular shoulders require undercuts.

If the model has a specified or modelled undercut, then determine r and t,

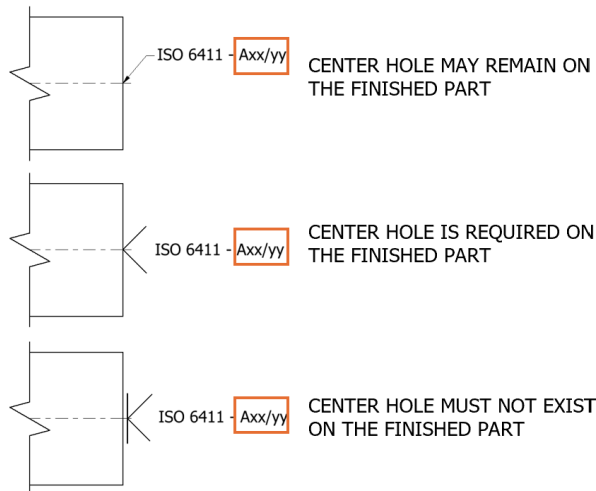


and determine the right notation in the form of DIN 509 - F r X t

Type	r^a $\pm 0,1$		t_1 +0,1 0	t_2 +0,05 0	f +0,2 0	g	Corresponding diameter d_1^b for workpieces	
	Series 1	Series 2					subjected to normal loads ^c	subjected to higher alternating loads
E		R0,2	0,1	—	1	—	Over Ø 1,6 up to Ø 3	—
	R0,4		0,2	—	2	—	Over Ø 3 up to Ø 18	—
		R0,6	0,2	—	2	—	Over Ø 10 up to Ø 18	—
		R0,6	0,3	—	2,5	—	Over Ø 18 up to Ø 80	—
	R0,8		0,3	—	2,5	—	Over Ø 18 up to Ø 80	—
		R1	0,2	—	2,5	—	—	Over Ø 18 up to Ø 50
		R1	0,4	—	4	—	Over Ø 80	—
	R1,2		0,2	—	2,5	—	—	Over Ø 18 up to Ø 50
	R1,2		0,4	—	4	—	Over Ø 80	—
	R1,6		0,3	—	4	—	—	Over Ø 50 up to Ø 80
	R2,5		0,4	—	5	—	—	Over Ø 80 up to Ø 125
	R4		0,5	—	7	—	—	Over Ø 125
F		R0,2	0,1	0,1	1	(0,9)	Over Ø 1,6 up to Ø 3	—
	R0,4		0,2	0,1	2	(1,1)	Over Ø 3 up to Ø 18	—
		R0,6	0,2	0,1	2	(1,4)	Over Ø 10 up to Ø 18	—
		R0,6	0,3	0,2	2,5	(2,1)	Over Ø 18 up to Ø 80	—
	R0,8		0,3	0,2	2,5	(2,3)	Over Ø 18 up to Ø 80	—
		R1	0,2	0,1	2,5	(1,8)	—	Over Ø 18 up to Ø 50
		R1	0,4	0,3	4	(3,2)	Over Ø 80	—
	R1,2		0,2	0,1	2,5	(2)	—	Over Ø 18 up to Ø 50
	R1,2		0,4	0,3	4	(3,4)	Over Ø 80	—
	R1,6		0,3	0,2	4	(3,1)	—	Over Ø 50 up to Ø 80
	R2,5		0,4	0,3	5	(4,8)	—	Over Ø 80 up to Ø 125
	R4		0,5	0,3	7	(6,4)	—	Over Ø 125
G	R0,4		0,2	0,2	(0,9)	(1,1)	Over Ø 3 up to Ø 18	—
H	R0,8		0,3	0,05	(2,0)	(1,1)	Over Ø 18 up to Ø 80	—
	R1,2		0,3	0,05	(2,4)	(1,5)	—	Over Ø 18 up to Ø 50
^a Undercuts with series 1 radii (as in DIN 250) are to be given preference. ^b Does not apply to parts with a short shoulder and thin-walled parts. When a workpiece has different diameters, it may be expedient for manufacturing reasons to use several undercuts having the same form and size. ^c Type G applies only to workpieces which are not subjected to high fatigue loads.								

Otherwise, use DIN 509 - F 0.8 × 0.3.

Center Holes



DESIGNATION	
DIAMETER	
5 TO 9	ISO 6411 – A 1.25/2.65
10 TO 20	ISO 6411 – A 2/4.25
21 TO 28	ISO 6411 – A 2.5/5.3
29 TO 50	ISO 6411 – A 5/10.6
51 TO 75	ISO 6411 – A 6.3/13.2
76 TO 100	ISO 6411 – A 8/17

☐ Correct designation on both sides of the shaft (center hole designation may be different on either side)

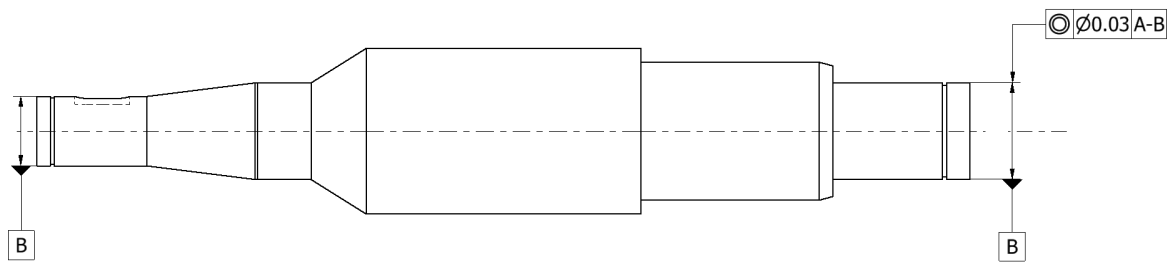
Secondary Tolerances

Shaft Length Dimensioning & Tolerancing

- pick a “reference datum” to dimension all lengths from
- with the exception of circlip positions, parts of shafts that house precision components like bearings need to have tighter tolerances (one step tighter than the general tolerance)
 - e.g. a 40mm long section of the shaft typically has $\pm 0.3\text{mm}$, but if holding a precision part, then the tolerance should be ± 0.2 (the next smaller “general” tolerance)
- ideally, all the dimensions should be measurable with a vernier caliper, and the total length dimension should be included
- do not over-dimension the lengths

Secondary Geometric Tolerances

- for the two ends of the shafts holding mechanical shaft fittings (like gears, bearings, etc.), they should be concentric to each other to a certain degree
- generally, a concentric tolerance of dia 0.03 is sufficient
- (maybe its only precision part shaft fittings?????????????)



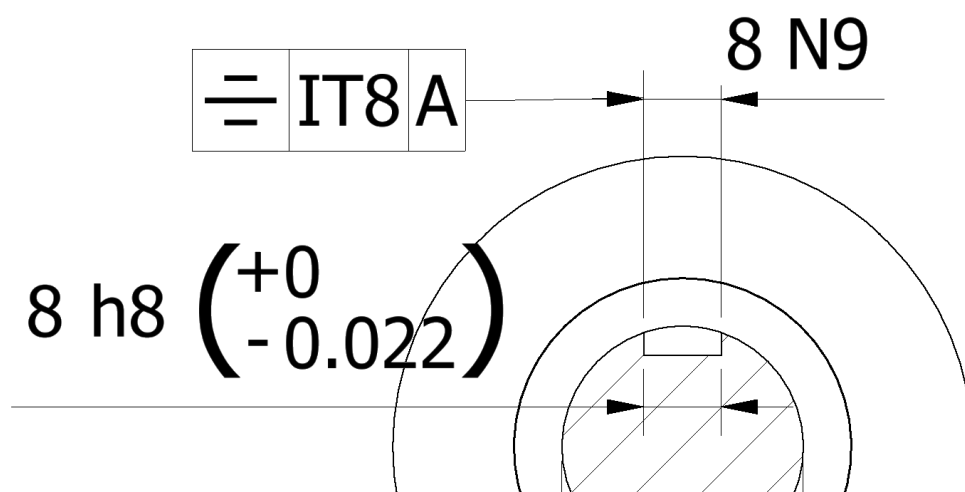
IT Geometric Tolerances

Calculating IT Value

For IT value X, controlling a dimension D, the IT value is the deviation for D hX.

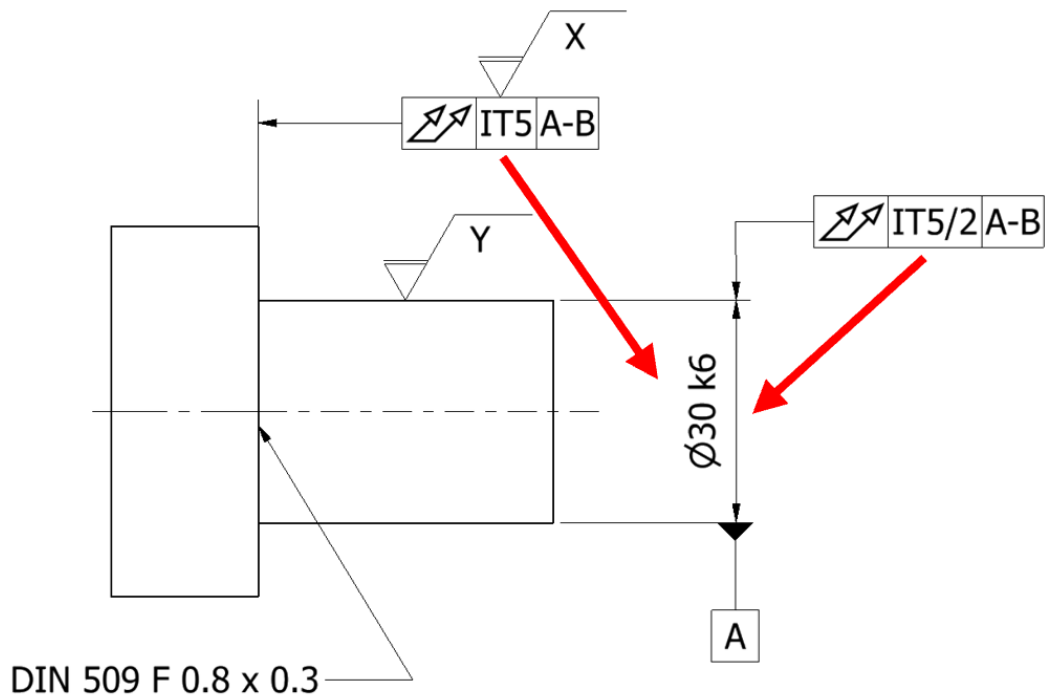
e.g. for IT8 geometric tolerance on a 8mm wide keyway, the deviation is 0.022.

Hence, replace IT8 with 0.022.



For bearing shoulders, the IT5 is based on the nominal diameter of the shaft.

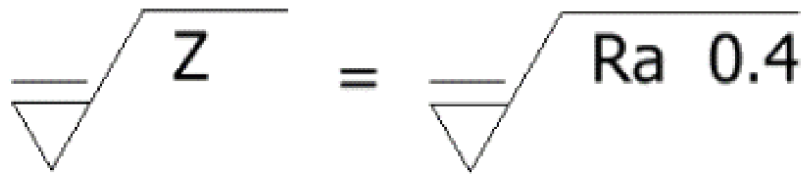
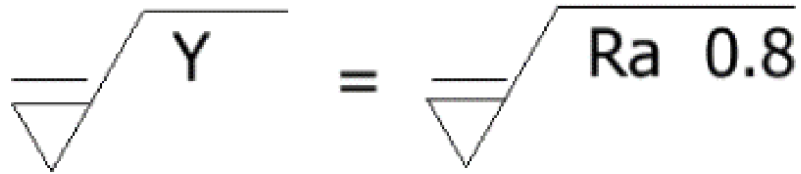
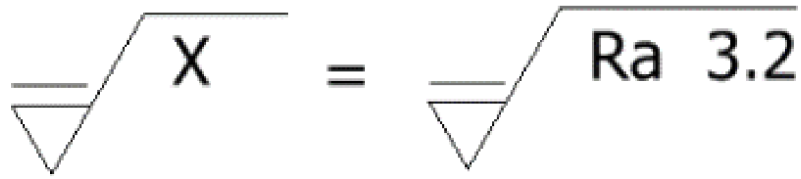
IT5/2 represents (IT5)/2, not IT(5/2).



Surface Finish

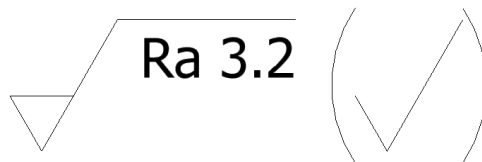
Surface Finish Convention

In the top right of the drawing,



Majority Surface Finish

The majority surface finish is usually Ra 3.2. (*****however Ra 1.6 may be feasible too***)



Annotations & Notes

☐ apply notes:

→ GENERAL NOTES:
ALL DIMENSIONS IN mm
GENERAL TOLERANCE:
FOR MACHINING ISO 2768-mK
ALL UNSPECIFIED CHAMFERS 1 X 45°
ALL CORNERS TO FOLLOW ISO 13715
CASE HARDENED TO HRc55±2
SURFACE TEXTURE IN ACCORDANCE WITH ISO 21920-1
SURFACE TEXTURES:

☐ apply centerlines

Other Useful things to note

General Clearance Fit: H7/g6

General Transitional Fit: H7/k6

General Interference/Press Fit: H7/p6

If two parts are fitted on the same section of the shaft, for each respective feature, the stricter one overrides the others.